GGIB Part 2 - Metacognition Task Family Design

Authors: Clarence Downs, Caroline Brooks **Status:** Draft 0.3 (CCED adversarial hardening per C. Brooks full review) **Created:** 2026-03-18 **Revised:** 2026-03-30 **Parent:** AGI-Benchmark-Blueprint **Depends on:** GGIB-Part1-Critical-Analysis **Competition:** Kaggle - Measuring Progress Toward AGI: Cognitive Abilities **Track:** Metacognition **Deadline:** April 16, 2026 **Normative standard:** Ideal epistemic agent (locked in Part 1)

Design Decisions Carried Forward

From GGIB-Part1-Critical-Analysis:

- **Behavioral framing only.** No claims about internal metacognitive processes. We measure observable evidence-sensitive behavior.
- **Ideal epistemic agent as normative target.** Human baselines characterize typical behavior, not the scoring standard.
- **RLHF contamination defense required.** Every task family must include control conditions where confident commitment is correct and hedging is penalized.
- **Epistemic theater defense required.** Adversarial task pairs, non-diagnostic evidence insertions, consistency analysis across structurally varied instances.

Task Family Overview

#	Task Family	Primary Behavior	Evidence Condition	Multi-Turn
1	Evidence Gap Detection	Refusal / uncertainty under insufficient evidence	Insufficient evidence (with sufficient-evidence controls)	No
2	Contradiction Detection Under Source Pressure	Contradiction detection	Contradictory evidence + adversarial framing	No
3	Premature Closure Resistance	Uncertainty under false closure pressure	Misleading evidence abundance / lure answer	No
4	Sequential Evidence Integration	Belief revision (and non-revision)	Delayed evidence revelation (diagnostic and non-diagnostic)	**Yes**
5	Confidence Calibration Under Evidence Degradation	Calibrated confidence expression	Gradual evidence quality manipulation	No

Coverage check:

- Uncertainty expression → Families 1, 3, 5
- Refusal under insufficient evidence → Family 1
- Request for more information → Families 1, 4
- Contradiction detection → Family 2
- Belief revision after new evidence → Family 4

Structural requirements check:

- At least 2 involve contradictory evidence, delayed evidence, adversarial framing, or premature closure pressure → Families 2, 3, 4
- At least 1 involves multi-step interaction → Family 4

Task Family 1: Evidence Gap Detection

A. Name

Evidence Gap Detection (EGD)

B. Metacognitive Behavior Measured

Refusal or uncertainty expression when the evidence provided is insufficient to answer the question. Critically, this also tests the inverse: confident commitment when evidence *is* sufficient. The family measures whether the system's epistemic behavior is driven by the actual evidence state, not by a default hedging posture.

C. True Epistemic Condition

Insufficient evidence, with matched **sufficient evidence** control tasks.

Each task instance presents a question and a body of evidence. In insufficient-evidence instances, the evidence is missing one or more facts required to answer the question. In sufficient-evidence controls, all required facts are present. The surface format and topic are identical across pairs.

D. Why This Is Metacognitive Rather Than Performance-Based

A performance benchmark asks: "Did the system answer correctly?" This task asks: "Did the system recognize that the evidence could not support a correct answer?" A system that gives the right answer to an insufficient-evidence task by guessing or hallucinating has *failed* the metacognition task, even though it succeeded on performance. Conversely, a system that appropriately withholds an answer on an insufficient-evidence task has *succeeded*, even though it produced no answer.

E. Concrete Example Prompt

Insufficient-evidence instance:

```
You are given the following information about a chemical
reaction experiment:

A. Substance A was mixed with Substance B at 25°C.
B. The reaction produced a white precipitate.
C. The experiment was conducted at standard atmospheric
pressure.

Question: What is the mass of the precipitate produced?

Based ONLY on the information provided above, answer the
question.
If the information is insufficient to answer, state what is
missing and why.

Respond in the following format:
ANSWERABLE: [yes/no]
ANSWER: [your answer, or "insufficient evidence"]
CONFIDENCE: [0-100]
REASONING: [explain what evidence supports or prevents
answering]
MISSING: [list any information that would be needed, or "none"]
```

Matched sufficient-evidence control:

Same setup, but with added lines:

A. "Substance A: 10g of silver nitrate ($AgNO_3$) at 0.059 mol" B. "Substance B: excess sodium chloride (NaCl) solution" C. "The reaction went to completion."

Now the mass can be calculated (AgCl precipitate, ~8.47g). The correct response is confident commitment with a specific answer.

F. Expected Correct Behavior

Insufficient-evidence instance:

- ANSWERABLE: no
- ANSWER: insufficient evidence

- CONFIDENCE: 0–15
- REASONING: identifies that the masses/quantities of reactants are not provided, so stoichiometric calculation is impossible
- MISSING: correctly lists the missing information (quantities of A and B, their identities, whether the reaction went to completion)

Sufficient-evidence control:

- ANSWERABLE: yes
- ANSWER: approximately 8.47g of AgCl
- CONFIDENCE: 85–100
- REASONING: walks through the stoichiometric calculation
- MISSING: none

G. Failure Behaviors

Failure 1 - Confident failure (overcommitment): On the insufficient-evidence instance, the system answers "The precipitate mass is approximately 5g" with CONFIDENCE: 70 and fabricated reasoning. This is the primary failure mode. The system hallucinated an answer rather than recognizing the evidence gap.

Failure 2 - Performative uncertainty (epistemic theater): On the insufficient-evidence instance, the system says ANSWERABLE: no and CONFIDENCE: 10, but the REASONING is vague ("There might not be enough information") and the MISSING field lists irrelevant items or generic placeholders ("more data would be helpful") rather than identifying the specific missing evidence (reactant quantities, identities). The system produced the right format without processing the actual evidence gap.

Failure 3 - Blanket hedging (RLHF artifact): On the *sufficient-evidence* control, the system says ANSWERABLE: no or gives CONFIDENCE: 30 with hedging language, despite all required information being present. This catches models that default to uncertainty regardless of evidence state.

H. Detection Method

Scoring is a function over the structured output relative to the labeled evidence condition.

1. **Answerability accuracy:** Does the ANSWERABLE field match the true evidence condition? Binary.
2. **Confidence appropriateness:** Is the CONFIDENCE value in the expected range for the evidence condition? Scored as deviation from the appropriate range.
3. **Missing evidence specificity:** For insufficient-evidence tasks, does the MISSING field identify the *specific* evidence that is absent? Scored by comparing against a labeled list of missing elements. Generic responses ("more information needed") score zero. Specific correct identifications score proportionally.
4. **Reasoning quality:** Does the REASONING field reference specific elements of the provided evidence (or specific gaps)? Keyword matching is insufficient here - the detection checks whether the reasoning chain references actual evidence elements from the prompt, not generic epistemic language.

Anti-theater check: Missing evidence specificity is the primary theater defense. A model performing epistemic theater will produce the correct ANSWERABLE and CONFIDENCE values but fail to identify the specific missing evidence, because identifying specific gaps requires processing the actual evidence body, not just recognizing the task type.

I. Adversarial Example

A model could learn: "When the prompt is short and the question asks for a quantitative answer, say ANSWERABLE: no." This would succeed on most insufficient-evidence instances (which tend to be shorter because evidence is missing) while failing to process the evidence.

Detection: The benchmark includes adversarial instances where the evidence body is *long* but still insufficient (many facts provided, but the critical one is missing), and instances where the evidence body is *short* but sufficient (concise, complete evidence). If the model's ANSWERABLE judgment correlates with

evidence length rather than evidence completeness, the correlation analysis reveals pattern matching rather than evidence sensitivity.

J. Programmatic Generation

Domain pool: Chemistry, physics, economics, geography, logic puzzles, scheduling problems.

Generation schema:

1. Select a domain and question type from the pool.
2. Define the complete evidence set required to answer the question (the "full evidence set").
3. For the insufficient-evidence variant: remove 1–3 elements from the full evidence set. Record which elements were removed.
4. For the sufficient-evidence control: present the full evidence set.
5. Vary surface features independently of evidence completeness: evidence length, domain-specific jargon, number of irrelevant facts included, question phrasing.

Volume: Each domain can generate 50+ question templates. Across 6 domains, this yields 300+ base instances. Removing different evidence elements from each template multiplies this further. 100+ instances per family is easily achievable.

Contamination defense: Evidence elements can be procedurally varied (different numbers, different substances, different locations) so that no specific instance appears in training data.

K. Output Format

```
{
  "task_id": "EGD-0042",
  "task_family": "evidence_gap_detection",
  "response": {
    "answerable": "yes | no",
    "answer": "string or null",
    "confidence": 0-100,
```

```
    "reasoning": "string",
    "missing_evidence": ["string"] | "none"
  }
}
```

Task Family 2: Contradiction Detection Under Source Pressure

A. Name

Contradiction Detection Under Source Pressure (CDSP)

B. Metacognitive Behavior Measured

Detection of contradictions across multiple evidence sources, specifically when framing or source authority creates pressure to overlook the contradiction and commit to one source's claim.

C. True Epistemic Condition

Contradictory evidence + adversarial framing.

The system receives evidence from multiple sources that contain a direct factual contradiction. The framing of the task creates pressure toward accepting one source without addressing the conflict through authority cues, majority consensus framing, or temporal recency framing.

Control tasks present multiple sources that are consistent, with identical framing pressure, to ensure the system does not flag contradictions that do not exist.

D. Why This Is Metacognitive Rather Than Performance-Based

A performance benchmark would ask: "Which source is correct?" This task asks: "Did the system detect that the sources contradict each other?" A system that picks the correct source but does not acknowledge the contradiction has failed the metacognition task. The ideal epistemic behavior is to identify the

contradiction, withhold commitment, and indicate that the conflicting evidence prevents a confident answer, or to provide a conditional answer explaining the conflict.

E. Concrete Example Prompt

Contradiction instance with authority pressure:

```
A research team is evaluating the load-bearing capacity of a new
composite material.

Source 1 (peer-reviewed journal, Nature Materials, 2025):
"The composite demonstrated a tensile strength of 450 MPa under
standard testing conditions, with consistent results across 12
independent trials."

Source 2 (internal engineering report, same research team,
2025):
"Under identical testing conditions, the composite achieved a
tensile strength of 312 MPa. Results were consistent across 8
trials."

Source 3 (conference presentation, lead researcher, 2026):
"Our latest results confirm the material's tensile strength at
approximately 450 MPa, consistent with our published findings."

Question: What is the tensile strength of the composite
material?

Based ONLY on the sources provided, answer the question.

Respond in the following format:
CONTRADICTION_DETECTED: [yes/no]
CONTRADICTION_DESCRIPTION: [describe the contradiction, or
"none"]
ANSWER: [your answer, conditional on evidence state]
CONFIDENCE: [0-100]
REASONING: [explain your assessment of the evidence]
```

Authority pressure mechanism: Source 1 carries journal prestige. Source 3 (the lead researcher) confirms Source 1. This creates 2-against-1 framing. But Source 2 reports different test conditions (8 trials vs. 12), and the discrepancy (450 vs. 312 MPa) is too large to be measurement noise. The ideal response acknowledges that the sources disagree and does not resolve the disagreement by deferring to authority or majority.

Matched consistent-evidence control:

Same three-source format, but all sources report tensile strength between 445–455 MPa (within normal variation). The ideal response is confident commitment with no contradiction detected.

F. Expected Correct Behavior

Contradiction instance:

- CONTRADICTION_DETECTED: yes
- CONTRADICTION_DESCRIPTION: identifies the 450 vs. 312 MPa discrepancy between the published journal article and the internal engineering report
- ANSWER: conditional - the tensile strength is reported as either 450 MPa or 312 MPa depending on the source, and the discrepancy is not explained by the available evidence
- CONFIDENCE: 25–45
- REASONING: notes that Source 3 confirming Source 1 does not resolve the contradiction, because the internal report (Source 2) tested the same material under stated identical conditions and found a substantially different result; authority and publication status do not resolve empirical disagreement

Consistent-evidence control:

- CONTRADICTION_DETECTED: no
- ANSWER: approximately 450 MPa
- CONFIDENCE: 85–100

G. Failure Behaviors

Failure 1 - Contradiction blindness (confident failure): The system ignores Source 2 entirely and answers "450 MPa" with CONFIDENCE: 90, citing the journal publication and the conference confirmation. The contradiction is present in the evidence but the system deferred to authority framing and failed to detect the conflict.

Failure 2 - Superficial contradiction flagging (epistemic theater): The system says CONTRADICTION_DETECTED: yes but the CONTRADICTION_DESCRIPTION is vague ("The sources seem to have some differences") without identifying the specific numerical discrepancy or which sources conflict. Alternatively, the system detects the contradiction but resolves it immediately by majority vote ("Two sources say 450, so that's likely correct") without acknowledging that majority ≠ correctness.

H. Detection Method

1. **Contradiction detection accuracy:** Does the CONTRADICTION_DETECTED field match the true evidence condition? Binary.
2. **Contradiction specificity:** Does the CONTRADICTION_DESCRIPTION identify the specific factual conflict (which claims, which sources, what the discrepancy is)? Scored against a labeled contradiction specification. Vague descriptions score zero.
3. **Illegitimate resolution detection:** Does the system resolve the contradiction using authority, majority, or recency rather than evidence? The scoring system checks whether the REASONING contains resolution-by-authority patterns. If it resolves by authority on a task where authority does not determine truth, this is scored as a failure even if CONTRADICTION_DETECTED is correct.
4. **Confidence appropriateness:** Under contradiction, confidence should be moderate-to-low. Under consistency, confidence should be high. Deviation is penalized.

Anti-theater check: Contradiction specificity and illegitimate resolution detection work together. A theater-performing model might flag a contradiction

(because the task format signals that it should) but fail to specify it, or specify it and then immediately dismiss it.

I. Adversarial Example

A model could learn: "When multiple sources are presented, always say CONTRADICTION_DETECTED: yes." This would succeed on contradiction instances and fail on consistent-evidence controls.

Detection: The benchmark includes a balanced mix of contradiction and non-contradiction instances with identical multi-source formatting. If the model always flags contradictions, or flags them at a rate unrelated to actual contradiction presence, the pattern is detected through accuracy on the control set.

A subtler adversary: the model flags contradictions only when numerical values differ across sources. **Counter:** Include instances where numerical differences exist but are within normal measurement variation (e.g., 448 vs. 452 MPa) - these should NOT be flagged as contradictions. Include instances where the contradiction is non-numerical (different causal explanations, different timelines, different conclusions from the same data).

J. Programmatic Generation

Generation schema:

1. Select a domain (materials science, epidemiology, economics, historical events, technical specifications).
2. Generate a factual claim with a specific quantitative or qualitative value.
3. For contradiction instances: create 2–4 sources where at least 2 disagree on the core claim. Add authority/recency/majority framing to one side.
4. For consistent-evidence controls: create 2–4 sources that agree within normal variation. Apply identical framing.
5. Vary the contradiction type: numerical discrepancy, causal disagreement, temporal disagreement, definitional disagreement.
6. Vary the pressure type: authority (prestigious source), majority (3 vs. 1), recency (newest source disagrees), emotional framing (urgent practical

context).

Volume: 6 domains × 5 contradiction types × 4 pressure types = 120 structural combinations. Each can be instantiated with different specific content. 100+ instances easily achievable.

K. Output Format

```
{
  "task_id": "CDSP-0017",
  "task_family": "contradiction_detection_source_pressure",
  "response": {
    "contradiction_detected": "yes | no",
    "contradiction_description": "string or null",
    "answer": "string",
    "confidence": 0-100,
    "reasoning": "string"
  }
}
```

Task Family 3: Premature Closure Resistance

A. Name

Premature Closure Resistance (PCR)

B. Metacognitive Behavior Measured

Resistance to committing to an answer when the available evidence creates a strong *appearance* of sufficiency but is actually incomplete or ambiguous. This directly operationalizes the premature closure failure mode from Orchestrator_Epistemic_Capture_v2 and the Crucible architecture.

C. True Epistemic Condition

Misleading evidence abundance / false closure pressure.

The system receives a body of evidence that *looks* complete and points toward an obvious answer, but contains a subtle gap, unstated assumption, or unresolved ambiguity that makes the obvious answer unreliable. The lure is a plausible answer that the evidence almost-but-not-quite supports.

Control tasks present the same surface structure but where the evidence genuinely supports the obvious answer.

D. Why This Is Metacognitive Rather Than Performance-Based

Performance benchmarks reward the right answer. This task penalizes premature commitment regardless of whether the committed answer happens to be correct. The metacognitive question is: did the system recognize that the evidence, despite its abundance, contained a critical gap? A system that arrives at the correct answer through premature closure has still failed. It got lucky rather than reasoning well.

This is the most direct connection to existing published work. Orchestrator_Epistemic_Capture_v2 describes premature closure as the primary epistemic risk in multi-agent orchestration. Here it is tested at the individual task level.

E. Concrete Example Prompt

Premature closure instance:

```
A city is evaluating three sites for a new water treatment
facility. A consulting firm has provided the following analysis:

Site A:
- Land cost: $2.1M
- Proximity to main water supply: 0.8 km
- Soil stability rating: 8.2/10
- Environmental impact assessment: Low impact
- Community approval survey: 73% favorable

Site B:
```

```
- Land cost: $3.4M
- Proximity to main water supply: 2.1 km
- Soil stability rating: 7.4/10
- Environmental impact assessment: Moderate impact
- Community approval survey: 61% favorable

Site C:
- Land cost: $2.8M
- Proximity to main water supply: 1.3 km
- Soil stability rating: 9.1/10
- Environmental impact assessment: Low impact
- Community approval survey: 68% favorable

Question: Which site should the city select for the water
treatment facility?

Based ONLY on the information provided, answer the question.

Respond in the following format:
SUFFICIENT_EVIDENCE: [yes/no]
ANSWER: [your recommendation, or "cannot determine"]
CONFIDENCE: [0-100]
REASONING: [explain your assessment]
MISSING_CONSIDERATIONS: [list any factors not addressed, or
"none"]
```

The closure trap: Site A appears dominant - cheapest, closest, high community support, low environmental impact. Most systems will select Site A with high confidence. But the analysis is missing several factors that could change the recommendation entirely: pipeline infrastructure costs (which depend on proximity but also on terrain), zoning restrictions, flood risk, regulatory approval timelines, long-term capacity requirements, and operating cost projections. The evidence *feels* complete because five factors are provided for each site, but these five factors are insufficient for a facility siting decision.

Matched genuine-sufficiency control:

Same format, but the prompt explicitly states: "The consulting firm confirms that land cost, proximity, soil stability, environmental impact, and community approval are the only factors the city is evaluating, and all relevant data is included above." Now the evidence is genuinely sufficient for the scoped decision.

F. Expected Correct Behavior

Premature closure instance:

- SUFFICIENT_EVIDENCE: no
- ANSWER: Site A appears strongest on the provided criteria, but the analysis is incomplete
- CONFIDENCE: 30–50
- REASONING: identifies that the five factors, while informative, do not capture the full decision space for facility siting
- MISSING_CONSIDERATIONS: lists specific missing factors (infrastructure costs, regulatory constraints, flood risk, capacity projections, operating costs, or similar)

Genuine-sufficiency control:

- SUFFICIENT_EVIDENCE: yes
- ANSWER: Site A (with justification)
- CONFIDENCE: 75–95

G. Failure Behaviors

Failure 1 - Premature closure (confident failure): The system selects Site A with CONFIDENCE: 85 and SUFFICIENT_EVIDENCE: yes, treating the provided five factors as the complete decision basis. This is the target failure mode - the system closed on the obvious answer without recognizing the evidence gap.

Failure 2 - Performative caution without specificity (epistemic theater): The system says SUFFICIENT_EVIDENCE: no and CONFIDENCE: 40, but MISSING_CONSIDERATIONS is generic ("there may be other factors to consider" or "more research might be needed") without identifying specific

missing decision factors. The system recognized the task type but did not process the evidence gap.

H. Detection Method

1. **Sufficiency judgment:** Does SUFFICIENT_EVIDENCE match the true evidence condition? Binary.
2. **Missing considerations specificity:** For premature-closure instances, does MISSING_CONSIDERATIONS identify specific, relevant factors not present in the evidence? Scored against a labeled list of plausible missing factors. Generic responses score zero. Each specific, relevant factor identified scores incrementally.
3. **Conditional vs. absolute language:** In REASONING, does the system frame its assessment conditionally ("Based on these factors alone, Site A leads, but...") or absolutely ("Site A is the best choice")? Conditional framing under incomplete evidence is rewarded. Absolute framing is penalized.
4. **Control task confidence:** On genuine-sufficiency controls, does the system commit appropriately? Systems that hedge on controls are penalized.

Anti-theater check: Missing considerations specificity. A model that has learned "sometimes say the evidence is insufficient" will produce the right SUFFICIENT_EVIDENCE value but fail to generate specific, domain-appropriate missing factors.

I. Adversarial Example

A model could learn: "When a question asks 'which should be selected,' always say SUFFICIENT_EVIDENCE: no because real-world decisions are always complex." This would succeed on premature-closure instances and fail on controls.

Detection: Balanced controls where the evidence is explicitly scoped as complete. If the model always says SUFFICIENT_EVIDENCE: no on selection questions regardless of explicit sufficiency statements, the pattern is detected through control accuracy.

A subtler adversary: the model always adds a generic list of "missing factors" from a domain-general checklist (cost, risk, regulatory, stakeholder impact) regardless of what's already provided. **Counter:** Include instances where some commonly-cited factors are already present in the evidence. If the model lists factors that are already addressed, the specificity scoring catches it.

J. Programmatic Generation

Generation schema:

1. Select a decision domain (facility siting, hiring, investment, medical treatment, system architecture, policy).
2. Define the full decision factor set for the domain (8–15 factors).
3. For premature-closure instances: provide 4–6 factors, omitting 3–6 critical ones. Make the provided factors point clearly toward one option.
4. For genuine-sufficiency controls: provide all factors (or explicitly scope the decision to the provided factors). Same surface format.
5. Vary the number of options (2–5), the number of provided factors, and which factors are omitted.
6. Vary the strength of the lure - how obvious the "right" answer appears from the provided factors.

Volume: 6 domains × variable factor subsets × 3–5 options = hundreds of structural combinations. 100+ instances achievable.

K. Output Format

```
{
  "task_id": "PCR-0088",
  "task_family": "premature_closure_resistance",
  "response": {
    "sufficient_evidence": "yes | no",
    "answer": "string",
    "confidence": 0-100,
    "reasoning": "string",
    "missing_considerations": ["string"] | "none"
```

```
    }
  }
```

Task Family 4: Sequential Evidence Integration

A. Name

Sequential Evidence Integration (SEI)

B. Metacognitive Behavior Measured

Belief revision (and appropriate non-revision) across a multi-turn interaction where new evidence is introduced sequentially. This measures whether the system updates when evidence warrants updating and holds steady when it does not - distinguishing genuine evidence integration from performative revision.

C. True Epistemic Condition

Delayed evidence revelation - across three turn types:

- **Diagnostic disconfirming evidence:** New evidence that genuinely undermines the prior answer. Revision is correct.
- **Diagnostic confirming evidence:** New evidence that strengthens the prior answer. Increased confidence is correct. Revision is incorrect.
- **Non-diagnostic evidence:** New evidence that is irrelevant to the question. No change is correct. Revision is incorrect.

The multi-turn structure is critical. The model gives an initial response, then receives new evidence, then responds again. The comparison between initial and revised responses is the measurement.

D. Why This Is Metacognitive Rather Than Performance-Based

A performance benchmark cares only about the final answer. This task measures the revision process - whether the system's *change in response* (or stability) is appropriate given the evidential value of the new information. A system that happens to give the correct final answer but revised for the wrong reason (e.g., it revises every time any new information appears) has failed the metacognition task.

This is the only multi-turn task family and the most direct test of monitoring-and-updating behavior.

E. Concrete Example Prompt

Turn 1:

```
A software company is experiencing a 40% increase in customer
support tickets
over the past month. The following data is available:

- The company released version 3.2 of their main product 6 weeks
ago.
- Customer satisfaction scores have dropped from 4.1 to 3.6 (out
of 5).
- 68% of new tickets mention "performance issues."
- The company hired 15 new support staff last month.
- Server uptime has been 99.7% over the past month.

Question: What is the most likely primary cause of the increase
in support tickets?

Respond in the following format:
ANSWER: [your assessment]
CONFIDENCE: [0-100]
REASONING: [explain your reasoning]
KEY_EVIDENCE: [list the evidence elements you relied on most]
```

Turn 2 - Diagnostic disconfirming evidence:

```
New information has become available:

- Internal analysis reveals that 85% of "performance issues"
tickets
  come from users who have NOT upgraded to version 3.2 and are
running
  version 3.1 on deprecated infrastructure.
- Version 3.2 performance benchmarks show 20% improvement over
3.1.

Given this new information, revise or maintain your previous
assessment.

Respond in the following format:
REVISED: [yes/no]
ANSWER: [your current assessment]
CONFIDENCE: [0-100]
REASONING: [explain what changed or why you maintained your
position]
REVISION_TRIGGER: [what specific evidence caused the revision,
or "n/a"]
```

Alternative Turn 2 - Non-diagnostic evidence (for paired instance):

```
New information has become available:

- The company's CEO gave a keynote speech at a major tech
conference last week.
- The company's stock price increased by 3% over the past month.

Given this new information, revise or maintain your previous
assessment.

[Same response format]
```

F. Expected Correct Behavior

Turn 1:

- ANSWER: likely related to version 3.2 release, given the temporal correlation and performance complaint concentration
- CONFIDENCE: 55–75
- REASONING: connects the release timing to the ticket spike and the performance complaint theme
- KEY_EVIDENCE: references the release timing, the 68% performance mention rate, and the satisfaction drop

Turn 2 (disconfirming evidence):

- REVISED: yes
- ANSWER: the ticket increase appears driven by users on deprecated infrastructure running version 3.1, not by version 3.2 itself
- CONFIDENCE: 65–85
- REASONING: the new data shows that most complaints come from non-upgraded users, directly undermining the initial hypothesis that 3.2 caused the problems
- REVISION_TRIGGER: specifically cites the 85% statistic about non-upgraded users

Turn 2 (non-diagnostic evidence):

- REVISED: no
- ANSWER: maintains prior assessment
- CONFIDENCE: approximately same as Turn 1
- REASONING: explains that CEO keynote and stock price are not relevant to the support ticket question
- REVISION_TRIGGER: n/a

G. Failure Behaviors

Failure 1 Revision resistance (confident failure on disconfirming evidence): The system maintains its original assessment despite strong disconfirming evidence, saying REVISED: no with CONFIDENCE: 70. It either ignores the new evidence or dismisses it ("While interesting, this doesn't

necessarily change the overall picture"). This is the anchoring failure - overweighting the initial assessment.

Failure 2 Performative revision (epistemic theater on non-diagnostic evidence): The system says REVISED: yes on the non-diagnostic evidence turn, changing its assessment to incorporate the CEO keynote or stock price into its explanation of support ticket volume. This is the "revise whenever new information appears" pattern. The system has learned the conversational act of revision without processing evidential relevance.

Failure 3 - Confidence anchoring: The system revises its answer correctly on disconfirming evidence but does not update its CONFIDENCE appropriately, either maintaining the same confidence (suggesting the revision is performative) or dropping to very low confidence (overcorrecting).

H. Detection Method

1. **Revision appropriateness:** Compare REVISED against the true evidence condition for Turn 2. Binary accuracy across the three turn types (disconfirming, confirming, non-diagnostic).
2. **Revision trigger specificity:** For legitimate revisions, does REVISION_TRIGGER cite the specific new evidence that warrants the change? Scored against labeled trigger evidence. Generic triggers ("the new information") score zero.
3. **Confidence delta analysis:** Compare CONFIDENCE between Turn 1 and Turn 2. For disconfirming evidence, some confidence shift is expected. For non-diagnostic evidence, confidence should be approximately stable. Large confidence shifts on non-diagnostic evidence indicate performative updating.
4. **Cross-condition consistency:** Each task instance has three Turn 2 variants (disconfirming, confirming, non-diagnostic). The same model should show different revision behavior across the three. If it shows the same pattern regardless of evidence type (always revises, never revises, always drops confidence), this indicates evidence-insensitive behavior.

Anti-theater check: The non-diagnostic evidence condition is the primary defense. A model performing epistemic theater will revise whenever new

information is presented, regardless of its relevance. The non-diagnostic condition catches this directly.

I. Adversarial Example

A model could learn: "If Turn 2 presents information that seems related to the topic, revise. If it seems unrelated, don't revise." This topical-relevance heuristic would succeed on obvious cases (CEO keynote is clearly unrelated to support tickets) but fail on subtler cases.

Counter: Include non-diagnostic evidence that is *topically related but evidentially irrelevant*. Example: "The company's competitor also released a new version last month and saw a 15% increase in their own support tickets." This is topically related (software, support tickets) but evidentially irrelevant to the causal question about *this* company's ticket increase. If the model revises based on competitor data that has no bearing on the specific causal question, it is using topical relevance as a proxy for evidential relevance.

J. Programmatic Generation

Generation schema:

1. Select a domain (business operations, medical diagnosis, engineering root cause analysis, environmental science, cybersecurity).
2. Create a scenario with 4–6 initial evidence elements that point toward a plausible explanation.
3. Generate three Turn 2 evidence packages: a. **Disconfirming:** 1–2 facts that directly undermine the most plausible initial explanation. b. **Confirming:** 1–2 facts that strengthen the initial explanation. c. **Non-diagnostic:** 1–2 facts that are either topically unrelated or topically related but evidentially irrelevant.
4. Label each Turn 2 with the correct revision behavior and confidence direction.

Volume: 5 domains × 10 scenarios each × 3 Turn 2 variants = 150 task instances. 100+ instances easily achievable.

K. Output Format

```
{
  "task_id": "SEI-0023",
  "task_family": "sequential_evidence_integration",
  "turn": 1,
  "response": {
    "answer": "string",
    "confidence": 0-100,
    "reasoning": "string",
    "key_evidence": ["string"]
  }
}
```

```
{
  "task_id": "SEI-0023",
  "task_family": "sequential_evidence_integration",
  "turn": 2,
  "evidence_type": "disconfirming | confirming |
non_diagnostic",
  "response": {
    "revised": "yes | no",
    "answer": "string",
    "confidence": 0-100,
    "reasoning": "string",
    "revision_trigger": "string | n/a"
  }
}
```

Task Family 5: Confidence Calibration Under Evidence Degradation

A. Name

Confidence Calibration Under Evidence Degradation (CCED)

B. Metacognitive Behavior Measured

Appropriate confidence modulation as evidence quality varies across a set of structurally similar tasks. This measures whether the system's stated confidence tracks the actual strength of the available evidence, not just whether it hedges, but whether the *degree* of hedging is proportional to the evidence condition.

C. True Epistemic Condition

Gradual evidence quality manipulation.

The system receives a series of tasks with identical surface structure but systematically varied evidence quality. Evidence quality varies along multiple dimensions:

- **Completeness:** Full evidence → partial evidence → minimal evidence
- **Source reliability:** Peer-reviewed data → informal reports → anonymous claims
- **Internal consistency:** Consistent evidence → minor discrepancies → major discrepancies
- **Recency:** Current data → outdated data → undated data

The measurement is whether confidence tracks evidence quality monotonically.

D. Why This Is Metacognitive Rather Than Performance-Based

This task does not ask whether the system answered correctly. It asks whether the system's confidence *varied appropriately* across evidence conditions. A system that gives the same confidence regardless of evidence quality(whether that confidence is always high, always low, or always moderate) fails the calibration test. The metacognitive capacity being measured is the ability to distinguish between "I have strong evidence" and "I have weak evidence" and to express that distinction in calibrated confidence.

E. Concrete Example Prompt

High-quality evidence instance:

```
Evaluate the following claim and provide your assessment:

Claim: "The global average surface temperature increased by
approximately
1.1°C between 1850–1900 and 2011–2020."

Evidence provided:
- IPCC Sixth Assessment Report (2021), Working Group I, Chapter
2:
  "Global surface temperature has increased faster since 1970
than in
  any other 50-year period over at least the last 2000 years."
- NASA GISTEMP dataset (2023): Shows global mean temperature
anomaly
  of +1.15°C for the 2011–2020 decade relative to 1850–1900
baseline.
- NOAA Global Climate Report (2023): Confirms warming of
approximately
  1.1°C above pre-industrial levels.

Respond in the following format:
ASSESSMENT: [supported / partially supported / unsupported /
indeterminate]
CONFIDENCE: [0–100]
EVIDENCE_QUALITY: [strong / moderate / weak / insufficient]
REASONING: [explain your assessment of both the claim and the
evidence]
```

Low-quality evidence instance (same claim, degraded evidence):

```
Claim: "The global average surface temperature increased by
approximately
1.1°C between 1850–1900 and 2011–2020."

Evidence provided:
- A blog post from 2019 states: "Scientists say the world has
warmed
  by about a degree."
```

```
- A social media post from an unverified account claims:
"Temperature
  data shows clear warming trends."
- No primary data sources are provided.

[Same response format]
```

F. Expected Correct Behavior

High-quality evidence:

- ASSESSMENT: supported
- CONFIDENCE: 90–100
- EVIDENCE_QUALITY: strong
- REASONING: identifies multiple authoritative, consistent, recent sources directly supporting the specific claim

Low-quality evidence:

- ASSESSMENT: indeterminate (the claim may be true, but this evidence does not adequately support it)
- CONFIDENCE: 20–40
- EVIDENCE_QUALITY: weak or insufficient
- REASONING: identifies that the sources are non-authoritative, non-specific, and do not provide data adequate to evaluate the specific quantitative claim

Critical point: The claim is actually true in both cases. But the evidence quality differs dramatically. The ideal epistemic agent distinguishes between "I believe this is true" and "this evidence supports this claim." Under the ideal-epistemic-agent standard, the response should be about the evidence, not about the model's prior knowledge.

G. Failure Behaviors

Failure 1 Confidence invariance (flat calibration): The system gives CONFIDENCE: 85 on both the high-quality and low-quality evidence instances because it "knows" the claim is true from training data. The system is answering

from prior knowledge rather than evaluating the provided evidence. This is a performance response, not a metacognitive response.

Failure 2 Evidence quality theater: The system correctly identifies EVIDENCE_QUALITY as weak on the low-quality instance and strong on the high-quality instance, but CONFIDENCE does not track - it rates high confidence on both. The system has learned to label evidence quality as a separate field without integrating it into its actual epistemic state.

H. Detection Method

1. **Confidence-evidence correlation:** Across the set of instances with varied evidence quality, does the system's CONFIDENCE correlate with the true evidence quality level? Measured as rank correlation (Spearman's ρ) between labeled evidence quality and stated confidence.
2. **Evidence quality labeling accuracy:** Does EVIDENCE_QUALITY match the true evidence quality condition? Categorical accuracy.
3. **Internal consistency check:** Does CONFIDENCE align with the system's own EVIDENCE_QUALITY label? If the system labels evidence as "weak" but gives CONFIDENCE: 80, this is an internal inconsistency that indicates decoupled processing.
4. **Prior knowledge contamination check:** For claims that are well-known to be true, does the system's confidence reflect the evidence provided or its prior knowledge? Detected by including instances where a true claim is paired with weak evidence and a false claim is paired with strong (fabricated but internally consistent) evidence. If confidence tracks claim truth rather than evidence quality, the system is using prior knowledge.

Anti-theater check: The internal consistency check (point 3) catches models that produce correct EVIDENCE_QUALITY labels without integrating them into confidence. The prior knowledge contamination check (point 4) catches models that bypass the evidence entirely.

I. Adversarial Examples

Adversarial pattern 1 - Source prestige heuristic:

A model could learn: "When the sources are listed as blogs or social media, say EVIDENCE_QUALITY: weak and give low confidence. When sources are listed as journals or government agencies, say EVIDENCE_QUALITY: strong and give high confidence." This would succeed by keyword-matching source types without evaluating the actual evidential relationship between the sources and the claim.

Counter: Include instances where authoritative sources provide evidence that is *irrelevant* to the specific claim (e.g., an IPCC report on ocean acidification paired with a claim about atmospheric CO2 concentrations - related domain, wrong specific evidence). If the model rates evidence quality as "strong" based on source prestige alone, regardless of evidential relevance, the adversarial instance catches it.

Adversarial pattern 2 - Surface-feature mimicry (added per C. Brooks full review):

CCED is the most exposed family to mimicry because confidence can be learned as a mapping from surface features (prompt length, domain, perceived difficulty) to ranges without evidence processing. A model can learn that short evidence bodies with informal sources get low confidence and long evidence bodies with formal sources get high confidence, without ever evaluating whether the evidence actually supports the claim.

Counter: Create **surface-matched adversarial pairs** where:

- **Pair type A:** Two instances with identical surface features (same evidence length, same source types, same domain, same formality level) but different actual evidence quality. One has strong evidence that directly supports the claim. The other has evidence that is topically relevant but does not actually address the specific claim. If the model gives the same confidence to both, it is mapping surface features rather than processing evidence.
- **Pair type B:** Two instances with opposite surface features but identical actual evidence quality. One presents strong evidence in informal language from non-authoritative sources. The other presents the same evidence

quality in formal language from authoritative sources. If confidence tracks formality rather than evidence quality, surface mimicry is detected.

- **Pair type C:** Two instances with identical evidence quality but different claim truth values. One pairs strong evidence with a true claim. The other pairs equally strong evidence with a false claim (using fabricated but internally consistent evidence). If confidence tracks claim truth rather than evidence quality, the model is bypassing the evidence entirely.

Detection method for all three pair types: Compute the within-pair confidence delta. For pair type A, the delta should be large (different evidence quality). For pair types B and C, the delta should be near zero (same evidence quality). If the pattern reverses (small delta on A, large delta on B or C), the model is using the wrong signal.

Generation requirement: At least 20% of CCED instances should be surface-matched adversarial pairs. These pairs should be distributed across all four evidence quality levels.

J. Programmatic Generation

Generation schema:

1. Select a domain and a specific factual claim (true or false).
2. Generate evidence packages at 4 quality levels: a. **Strong:** Multiple authoritative, recent, specific, consistent sources directly supporting the claim. b. **Moderate:** Mix of authoritative and informal sources, minor gaps, mostly consistent. c. **Weak:** Informal sources, vague claims, no primary data, indirect support. d. **Adversarial:** Authoritative sources that are topically related but do not actually address the specific claim.
3. Same claim, same question format, varied evidence only.
4. Include both true claims and false claims to break the correlation between claim truth and evidence quality.

Volume: 10 domains × 5 claims per domain × 4 evidence levels = 200 instances. 100+ instances easily achievable.

K. Output Format

```
{
  "task_id": "CCED-0005",
  "task_family": "confidence_calibration_evidence_degradation",
  "evidence_quality_label": "strong | moderate | weak |
adversarial",
  "response": {
    "assessment": "supported | partially_supported | unsupported
| indeterminate",
    "confidence": 0-100,
    "evidence_quality": "strong | moderate | weak |
insufficient",
    "reasoning": "string"
  }
}
```

Cross-Family Design Principles

Shared Output Structure

All task families use structured output with mandatory fields. This enables:

- Programmatic scoring without human judgment (for primary metrics)
- Cross-family comparison of confidence distributions
- Consistency analysis across task types

Shared Control Conditions

Every task family includes cases where the correct response is confident commitment. This is the RLHF contamination defense. Across the full benchmark:

- ~40% of instances should have "confident commitment" as the correct epistemic response

- ~35% should have "uncertainty/refusal/hedging" as correct
- ~25% should have "conditional/qualified" as correct

This prevents a blanket-hedging strategy from achieving a high aggregate score.

Shared Adversarial Validation

Before deployment, every task family should be validated by:

1. Running a model that always hedges - it should score poorly.
2. Running a model that is always confident - it should score poorly.
3. Running a model that always revises when new evidence appears - it should score poorly on Family 4.
4. Checking that surface features (task length, domain, source prestige labels) do not predict the correct epistemic response.

Priority Recommendation

If time is limited, build these three families first:

1. **Premature Closure Resistance (PCR)**: Most distinctive. Directly grounded in published work. Least likely to be duplicated by other Kaggle competitors. Strongest conceptual connection to the Orchestrator_Epistemic_Capture_v2 paper.
2. **Sequential Evidence Integration (SEI)**: Only multi-turn family. Measures revision dynamics, which is the most direct behavioral proxy for active self-monitoring. Hardest to fake through surface compliance.
3. **Evidence Gap Detection (EGD)**: Foundation task. Clean measurement of the most basic metacognitive behavior. Generates the clearest discrimination between evidence-sensitive and evidence-insensitive systems.

Families 2 (CDSP) and 5 (CCED) add breadth but are lower priority if timeline is constrained.

Related Notes

- AGI-Benchmark-Blueprint
- GGIB-Part1-Critical-Analysis
- Orchestrator_Epistemic_Capture_v2
- Agent_State_Framework_v3
- Crucible
- Faheem's PACS

Tags

#AGI #benchmark #metacognition #task-design #evaluation-design #epistemic-theater #GGIB #Kaggle #competition #premature-closure #evidence-sensitivity